

Tapas Pal

Group Leader | Secure Computation & Communication, KASTEL SRL,
Karlsruhe Institute of Technology (KIT)

Ph.D. in Cryptography, IIT Kharagpur

📍 Am Fasanengarten 5, 76131 Karlsruhe, Germany

📞 (+49) 15510083039 ✉️ tapas.pal@kit.edu

🌐 tapaspal9.github.io/homepage 🆔 0000-0001-6278-0418

🎓 Google Scholar DBLP IACR CryptoDB



RESEARCH PROFILE

Cryptographer working on the foundations and real-world deployment of advanced public-key encryption. My research designs provably secure *functional*, *attribute-based*, and *registered* encryption schemes that decentralize trust, make computation on encrypted data *verifiable*, and bring strong privacy guarantees to cloud platforms, digital infrastructure, and AI systems. The work spans rigorous theory (pairing- and lattice-based constructions with formal security proofs) and applied impact (multiple international patents and industry technology due-diligence). Author of 20+ peer-reviewed papers at leading venues (CRYPTO, EUROCRYPT, ASIACRYPT, PKC, Journal of Cryptology).

Research Interests Functional & attribute-based encryption; registered and multi-authority encryption; verifiable computation on encrypted data; privacy-enhancing technologies; lattice- and pairing-based cryptography; witness encryption & obfuscation; threshold and post-quantum cryptography.

EDUCATION

Ph.D. in Cryptography, Indian Institute of Technology Kharagpur, India 2016 – 2021
Advisor: Prof. Ratna Dutta. Thesis: *Designing Provably Secure Advanced Cryptographic Primitives — Witness Encryption, Fully Homomorphic Encryption, and Functional Encryption*. Defended 6 October 2021.

M.Sc. in Mathematics, Indian Institute of Technology Kharagpur, India 2014 – 2016
CGPA 9.47/10 · Batch Rank 2.

B.Sc. (Honours) in Mathematics, Scottish Church College, University of Calcutta, India 2011 – 2014
First Class First · 88.88%.

ACADEMIC & RESEARCH APPOINTMENTS

Group Leader, Secure Computation & Communication, KASTEL Security Research Labs, KIT, Germany 2024 – present
Lead a research group on privacy-enhancing technologies; research strategy, proposal coordination, and mentoring of early-career researchers.

Postdoctoral Fellow, Social Informatics Laboratories, NTT Corporation, Tokyo, Japan 2022 – 2023
Computing on encrypted data and advanced functional-encryption primitives.

Visiting Researcher, Cyber Security Group, University of St. Gallen, Switzerland 2023: Sep-Oct
Supervised the project on multi-client attribute-based unbounded inner-product functional encryption.

Visiting Scientist, R. C. Bose Centre for Cryptology & Security, ISI Kolkata, India 2021: Nov-Dec
Functional encryption and its applications.

Research Intern, Cryptography & Information Security (CIS) Lab, NTT Research, Inc., Sunnyvale, USA 2021: Feb-Oct
Functional encryption.

Junior & Senior Research Fellow, Department of Mathematics, IIT Kharagpur, India 2016 – 2021
Doctoral research in cryptography.

RESEARCH FUNDING & GRANTS

Swiss National Science Foundation (SNSF), University of St. Gallen 2023
Multi-Client Unbounded Attribute-Based Inner Product Functional Encryption — with PI Prof. Katerina Mitrokotsa. (Completed.)

SELECTED HONOURS & AWARDS

Competing position offers (declined)

2023

Senior Researcher, Cyber Security Group, University of St. Gallen, Switzerland; and Senior Scientist / Lecturer, Cyber Security Research Group, University of Klagenfurt, Austria.

ERCIM Postdoctoral Fellowship, NTNU, Norway

2021

European Research Consortium for Informatics and Mathematics.

Junior Research Fellowship (JRF), UGC, Mathematical Sciences

2016

All-India Rank 78.

Joint Admission Test for M.Sc. (JAM), Mathematics

2014

All-India Rank 121.

Seven merit medals & prizes for excellence in Mathematics, Scottish Church College

2013 – 2014

Including the Krishnalal De Medal (best performance in Mathematics), Pravat Kr. Ghosh Medal, Prof. Alexander Thomson Prize, and Dr. Alexander Duff Memorial Prize.

PUBLICATIONS

In theoretical cryptography, author names are listed *alphabetically* by convention; my name is shown in **bold**. Full list: [Google Scholar](#) · [DBLP](#) · [IACR CryptoDB](#).

Bibliometrics: Citations 205 · h-index 09 (Google Scholar, June 11, 2026).

Peer-Reviewed Conference Proceedings

- [C1] **CRYPTO 2026**. Shalini Banerjee, **Tapas Pal**, Andy Rupp, Daniel Slamanig. *Simple Public-Key Anamorphic Encryption and Signature using Multi-Message Extensions*. (To appear.) [\[ePrint\]](#)
- [C2] **PKC 2026**. **Tapas Pal**, Robert Schädlich. *Registered Functional Encryption for Attribute-Weighted Sums with Access Control*. 29th Intl. Conf. on Practice and Theory of Public-Key Cryptography. [\[Proceedings\]](#) [\[ePrint\]](#)
- [C3] **ASIACRYPT 2025**. **Tapas Pal**, Robert Schädlich. *A General Framework for Registered Functional Encryption via User-Specific Pre-Constraining*. Advances in Cryptology — ASIACRYPT 2025. [\[Proceedings\]](#) [\[ePrint\]](#)
- [C4] **PKC 2025**. Subhranil Dutta, Aikaterini Mitrokotsa, **Tapas Pal**, Tomy Jenit. *Multi-Client Attribute-Based Unbounded Inner Product Functional Encryption, and More*. [\[Proceedings\]](#) [\[ePrint\]](#)
- [C5] **ASIACRYPT 2024**. Pratish Datta, **Tapas Pal**, Shota Yamada. *Registered FE beyond Predicates: (Attribute-Based) Linear Functions and More*. Advances in Cryptology — ASIACRYPT 2024. [\[Proceedings\]](#) [\[ePrint\]](#)
- [C6] **IEEE EuroS&P 2024**. Uddipana Dowerah, Subhranil Dutta, Frank Hartmann, Aikaterini Mitrokotsa, Sayantan Mukherjee, **Tapas Pal**. *SACfe: Secure Access Control in Functional Encryption with Unbounded Data*. [\[Proceedings\]](#) [\[ePrint\]](#)
- [C7] **EUROCRYPT 2024**. Taiga Hiroka, Fuyuki Kitagawa, Tomoyuki Morimae, Ryo Nishimaki, **Tapas Pal**, Takashi Yamakawa. *Certified Everlasting Secure Collusion-Resistant Functional Encryption, and More*. Advances in Cryptology — EUROCRYPT 2024. [\[Proceedings\]](#) [\[ePrint\]](#)
- [C8] **PKC 2023**. Pratish Datta, **Tapas Pal**. *Decentralized Multi-Authority Attribute-Based Inner-Product FE: Large Universe and Unbounded*. 26th Intl. Conf. on Practice and Theory of Public-Key Cryptography. [\[Proceedings\]](#) [\[ePrint\]](#)
- [C9] **ASIACRYPT 2022**. Pratish Datta, **Tapas Pal**, Katsuyuki Takashima. *Compact FE for Unbounded Attribute-Weighted Sums for Logspace from SXDH*. Advances in Cryptology — ASIACRYPT 2022. [\[Proceedings\]](#) [\[ePrint\]](#)
- [C10] **ASIACRYPT 2021**. Pratish Datta, **Tapas Pal**. *Compact Adaptively Secure FE for Attribute-Weighted Sums from k -Lin*. Advances in Cryptology — ASIACRYPT 2021. [\[Proceedings\]](#) [\[ePrint\]](#)
- [C11] **ProvSec 2021**. Subhranil Dutta, **Tapas Pal**, Ratna Dutta. *Fully Secure Unbounded Zero Inner Product Encryption with Short Ciphertexts and Keys*. 15th Intl. Conf. on Provable and Practical Security. [\[Proceedings\]](#)
- [C12] **ACISP 2021**. **Tapas Pal**, Ratna Dutta. *Chosen-Ciphertext Secure Functional Encryption from Constrained Witness PRF*. 26th Australasian Conf. on Information Security and Privacy. [\[Proceedings\]](#) [\[ePrint\]](#)
- [C13] **ACISP 2021**. **Tapas Pal**, Ratna Dutta. *CCA-Secure Attribute-Hiding Inner Product Encryption from Minimal Assumption*. 26th Australasian Conf. on Information Security and Privacy. [\[Proceedings\]](#) [\[ePrint\]](#)
- [C14] **LATINCRYPT 2021**. **Tapas Pal**, Ratna Dutta. *Attribute-Based Access Control for Inner Product Functional Encryption from LWE*. 7th Intl. Conf. on Cryptology and Information Security in Latin America. [\[Proceedings\]](#) [\[ePrint\]](#)
- [C15] **CANS 2020**. **Tapas Pal**, Ratna Dutta. *Chosen-Ciphertext Secure Multi-Identity and Multi-Attribute Pure FHE*. 19th Intl. Conf. on Cryptology and Network Security. [\[Proceedings\]](#) [\[ePrint\]](#)
- [C16] **ProvSec 2020**. **Tapas Pal**, Ratna Dutta. *Semi-Adaptively Secure Offline Witness Encryption from Puncturable Witness PRF*. 14th Intl. Conf. on Provable and Practical Security. [\[Proceedings\]](#) [\[ePrint\]](#)
- [C17] **ACISP 2019**. **Tapas Pal**, Ratna Dutta. *Offline Witness Encryption from Witness PRF and Randomized Encoding in CRS Model*. 24th Australasian Conf. on Information Security and Privacy. [\[Proceedings\]](#) [\[ePrint\]](#)

Journal Articles

- [J1] **IACR CiC 2025**. Subhranil Dutta, **Tapas Pal**, Amit Kumar Singh, Sourav Mukhopadhyay. *Fully Collusion Resistant Traceable Identity-Based Inner Product Functional Encryption*. IACR Communications in Cryptology. [\[Journal\]](#)

- [J2] **DCC 2024**. Pratish Datta, **Tapas Pal**, Katsuyuki Takashima. *Compact FE for Unbounded Attribute-Weighted Sums for Logspace from SXDH*. Designs, Codes and Cryptography. [Journal]
- [J3] **TCS 2024**. Subhranil Dutta, **Tapas Pal**, Ratna Dutta. *Reinforcing Privacy in Cloud Computing via Adaptively Secure Non-Zero Inner Product Encryption and Anonymous Identity-Based Revocation in the Unbounded Setting*. Theoretical Computer Science, vol. 995. [Journal]
- [J4] **JoC 2023**. Uddipana Dowerah, Subhranil Dutta, Aikaterini Mitrokotsa, Sayantan Mukherjee, **Tapas Pal**. *Unbounded Predicate Inner Product Functional Encryption from Pairings*. Journal of Cryptology, 36(29). [Journal] [ePrint]
- [J5] **DCC 2023**. Pratish Datta, **Tapas Pal**. *Compact Adaptively Secure FE for Attribute-Weighted Sums from k -Lin*. Designs, Codes and Cryptography, 91, 2917–3034. [Journal] [ePrint]

Manuscripts under Submission / Preprints

- [P1] **Preprint 2025**. **Tapas Pal**, Robert Schädlich, Erkan Tairi. *Registered Functional Encryption for Pseudorandom Functionalities from Lattices: Registered ABE for Unbounded-Depth Circuits and Turing Machines, and More*. [ePrint]

PATENTS

- [PT1] **US 2024**. Pratish Datta, **Tapas Pal**. *Decentralized Multi-Authority Attribute-Based Encryption for Large Universe and Unbounded*. US Patent WO2024151871A1. [Google Patents]
- [PT2] **US 2024**. Pratish Datta, **Tapas Pal**. *Compact Functional Encryption for Unbounded Attribute-Weighted Sums*. US Patent WO2024098074A2. [Google Patents]
- [PT3] **US/JP/EPO 2023**. Pratish Datta, Monosij Maitra, **Tapas Pal**. *Decentralized Multi-Authority Attribute-Based Inner-Product Functional Encryption*. Patent EP4254858A1. [Google Patents]
- [PT4] **US 2023**. Pratish Datta, **Tapas Pal**. *Compact Adaptively Secure Functional Encryption for Attribute-Weighted Sums*. US Patent WO2023014969A1. [Google Patents]

INVITED TALKS & RESEARCH VISITS

Invited Talks

- Swiss Crypto Day**, ETH Zurich, Switzerland Sep 2023
Inner Product Functional Encryption for the Real World.
- Research Seminar**, University of Auckland, New Zealand Jun 2019
From Functional Encryption to Indistinguishability Obfuscation.

Research Visits

- University of St. Gallen, Switzerland**, Cyber Security Group 2023
Host: Prof. Katerina Mitrokotsa.
- IIT Madras, India** 2019
Host: Prof. Shweta Agrawal.
- University of Auckland, New Zealand** 2019
Host: Prof. Steven D. Galbraith.

TEACHING EXPERIENCE

- Cryptography and Network Security — Teaching Assistant**, NPTEL / IIT Kharagpur 2018 – 2020
Spring 2018, 2019, 2020.
- Introduction to Abstract and Linear Algebra — Teaching Assistant**, NPTEL / IIT Kharagpur 2018 – 2020
Autumn 2018, 2019, 2020.
- Mathematics-II — Teaching Assistant**, IIT Kharagpur 2019
Spring 2019.

PROFESSIONAL SERVICE

- Program Committee** PKC 2027; ACM CCS 2024, 2026; WWW 2026; INDOCRYPT 2024, 2025; ICSP 2024, 2025.
- External Reviewer** CRYPTO 2022–2026; EUROCRYPT 2024–2026; ASIACRYPT 2024, 2025; TCC 2022, 2025; PKC 2023, 2024; ACNS 2022; AsiaCCS 2022; INDOCRYPT 2020, 2021.
- Journal Referee** Journal of Cryptographic Engineering; Designs, Codes and Cryptography; IEEE Trans. Information Forensics & Security; IEEE Trans. Information Theory; IET Information Security; Advances in Mathematics of Communications; Computer Standards & Interfaces.
- Expert Reviewer** American Mathematical Society (AMS): Mathematical Reviews Database

INDUSTRY CONSULTING & KNOWLEDGE TRANSFER

Cryptography Consultant (pro bono) , D11Z.Ventures GmbH & Co. KG	Nov 2025
Technical due-diligence on a password-less access-control platform (MyCena); startup evaluation and advisory.	
Cryptography Consultant (pro bono) , D11Z.Ventures GmbH & Co. KG	Mar 2025
Technical due-diligence on cryptographic image-authenticity technology (TrustNXT); startup evaluation and advisory.	

TECHNICAL SKILLS & LANGUAGES

Programming	C/C++.
Scientific Computing	MATLAB, Mathematica.
Tools & Typesetting	$\text{\LaTeX}$, Linux.
Languages	Bengali (native); Hindi (fluent); English (fluent, full professional working proficiency).

REFERENCES

Prof. Ratna Dutta

Dept. of Mathematics, IIT Kharagpur, India
ratna@maths.iitkgp.ac.in
facweb.iitkgp.ac.in/~ratna

Dr. Andy Rupp

Faculty of Science, Technology & Medicine,
University of Luxembourg, Luxembourg
andy.rupp@uni.lu
uni.lu/.../andy-rupp

Dr. Ryo Nishimaki

Distinguished Research Scientist,
Social Informatics Laboratories, NTT, Tokyo, Japan
ryo.nishimaki@ntt.com
nishimaki.info

Prof. Sourav Mukhopadhyay

Dept. of Mathematics, IIT Kharagpur, India
sourav@maths.iitkgp.ac.in
facweb.iitkgp.ac.in/~sourav

Prof. Katerina Mitrokotsa

School of Computer Science,
University of St. Gallen, Switzerland
katerina.mitrokotsa@unisg.ch
cybersecurity.unisg.ch/people/Katerina